Tutorial-2

CSL 100 / MAL 400

Recap

- Flow to write and run C programs.
- Data types
  - Integer, Float, Double, Char
- Printf
  - Printing sentences
  - Printing values
- Scanf
  - Taking input from keyboard

I/O Multiple Values

```
#include <stdio.h>
int main()
{
    int a;
    float b;
    printf("Enter integer and then a float: ");
    scanf("%d %f", &a, &b);
    printf("You entered %d and %f", a, b);
    return 0;
}
```

Centigrade to Fahrenheit

```
#include <stdio.h>
int main()
{    float cent, fahr;
      printf(“enter centigrade value”);
      scanf(“%f”,&cent);
      fahr = cent*(9.0/5.0) + 32;
      printf( “%f C equals %f F\n”, cent, fahr);
      return 0;
}
```

ASCII TABLE

Dec	Char	Dec	Char	Dec	Char	Dec	Char	Dec	Char	Dec	Char
...	...	48	0	64	@	80	P	96	`	112	p
33	!	49	1	65	A	81	Q	97	a	113	q
34	"	50	2	66	B	82	R	98	b	114	r
35	#	51	3	67	C	83	S	99	c	115	s
36	$	52	4	68	D	84	T	100	d	116	t
37	%	53	5	69	E	85	U	101	e	117	u
38	&	54	6	70	F	86	V	102	f	118	v
39	'	55	7	71	G	87	W	103	g	119	w
40	(	56	8	72	H	88	X	104	h	120	x
41	)	57	9	73	I	89	Y	105	i	121	y
42	*	58	:	74	J	90	Z	106	j	122	z
43	+	59	;	75	K	91	[	107	k	123	{
44	,	60	<	76	L	92	\	108	l	124	|
45	-	61	=	77	M	93	]	109	m	125	}
46	.	62	>	78	N	94	^	110	n	126	~
47	/	63	?	79	O	95	_	111	o	127	DEL

ASCII Value To Character

```
#include <stdio.h>
int main()
{
    char chr;
    printf("Enter a character: ");
    scanf("%c", &chr);
    printf("You entered %c.\n",chr);

    printf("ASCII value is %d.", chr);
    return 0;
}
```

Try printing the value of integer but give a character as an input using scanf.

Precision and Formatting in C

Format Specifier	Type
%c	Used to print a character
%d	Used to print the signed integer
%f	Used to print the float values
%i	Used to print the unsigned integer
%l	Used to print the long integer
%lf	Used to print the double values
%lu	Used to print the unsigned integer or unsigned long integer
%s	Used to print the string
%u	Used to print the unsigned integer

scanf ("%2d%3d", &a, &b);

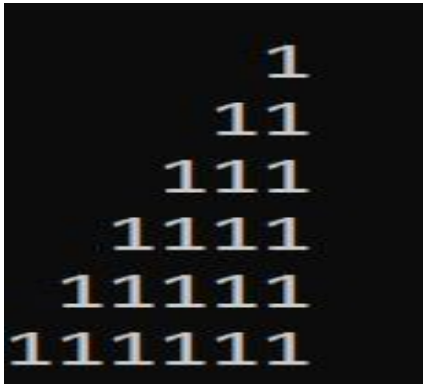
printf("a=%3d, b=%4d", a, b);

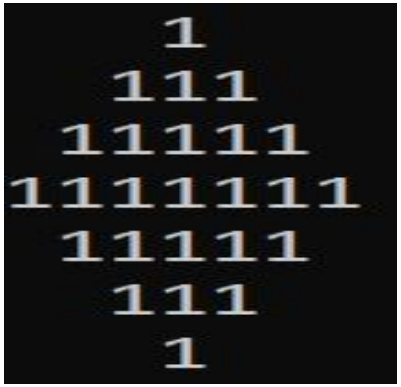
scanf ("%3f%3f", &a, &b);

printf("x = %4.1f, y = %7.2f ", x, y);

scanf ("%d %*d %d", &a, &b, &c);

Printing Patterns





- **printf("%5s", ""); // this helps us in printing spaces used for**
  // indentation
- **printf("%3d",11); // such formatting can also be used**

Quiz 2

- Write a C program that reads a character from the user and prints its ASCII value.
- Write a C program that converts a lowercase character entered by the user to its uppercase equivalent.
- Write a C program that reads two characters and prints the difference in their ASCII values.